1.10 Vectors				
1.10a 1.10b	Vectors	a) Be able to use vectors in two dimensions. *i.e. Learners should be able to use vectors expressed as* $x\mathbf{i}+y\mathbf{j}$ *or as a column vector* $\begin{pmatrix} x \\ y \end{pmatrix}$, *to use vector notation appropriately either as* $\overrightarrow{AB}$ *or* $\mathbf{a}$. *Learners should know the difference between a scalar and a vector, and should distinguish between them carefully when writing by hand.*	b) Be able to use vectors in three dimensions. *i.e. Learners should be able to use vectors expressed as* $x\mathbf{i}+y\mathbf{j}+z\mathbf{k}$ *or as a column vector* $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$. *Includes extending 1.10c to 1.10g to include vectors in three dimensions, excluding the direction of a vector in three dimensions.*	MJ1
1.10c	Magnitude and direction of vectors	c) Be able to calculate the magnitude and direction of a vector and convert between component form and magnitude/direction form. *Learners should know that the modulus of a vector is its magnitude and the direction of a vector is given by the angle the vector makes with a horizontal line parallel to the positive x-axis. The direction of a vector will be taken to be in the interval* $[0°, 360°)$. *Includes use of the notation* $\lvert\mathbf{a}\rvert$ *for the magnitude of* $\mathbf{a}$ *and* $\lvert\overrightarrow{OA}\rvert$ *for the magnitude of* $\overrightarrow{OA}$. *Learners should be able to calculate the magnitude of a*		MJ2

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
1.10d	Basic operations on vectors	d) Be able to add vectors diagrammatically and perform the algebraic operations of vector addition and multiplication by scalars, and understand their geometrical interpretations. *i.e. Either a scaling of a single vector or a displacement from one position to another by adding one or more vectors, often in the form of a triangle of vectors.*		MJ3
1.10e 1.10f	Position vectors Distance between points	e) Understand and be able to use position vectors. *Learners should understand the meaning of displacement vector, component vector, resultant vector, parallel vector, equal vector and unit vector.* f) Be able to calculate the distance between two points represented by position vectors. *i.e. The distance between the points* $a\mathbf{i}+b\mathbf{j}$ *and* $c\mathbf{i}+d\mathbf{j}$ *is* $\sqrt{(c-a)^2+(d-b)^2}$.		MJ4
1.10g 1.10h	Problem solving using vectors	g) Be able to use vectors to solve problems in pure mathematics and in context, including forces.	h) Be able to use vectors to solve problems in kinematics. *e.g. The equations of uniform acceleration may be used in vector form to find an unknown. See section 3.02e.*	MJ5